CMR TECHNICAL CAMPUS
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Department of Computer Science and Engineering

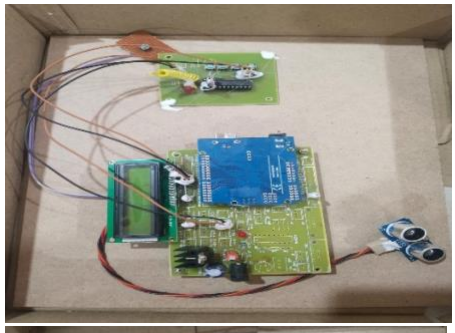
DESIGNING THE HAPTIC INTERFACE SYSTEM FOR THE DISABLED INDIVIDUALS

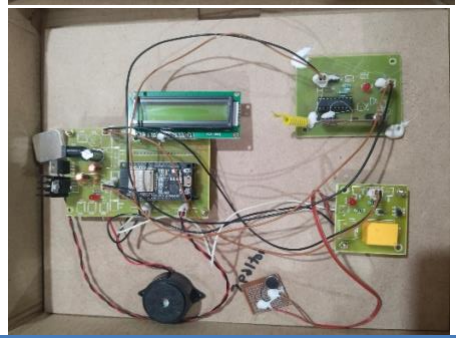
ABSTRACT

The haptic interface system is designed to help disabled individuals interact with their surroundings using tactile feedback instead of only visual or audio cues. The system uses wearable devices with sensors to detect obstacles and motion, and processes this data using microcontrollers such as ESP32 or Arduino. The processed information is converted into vibration patterns that guide the user during navigation and daily activities. The design focuses on affordability, portability, and low power consumption to improve safety and independence.

DESCRIPTION

The haptic interface system is designed to assist disabled individuals by providing touch-based feedback as an alternative to traditional visual and audio interfaces. The system uses a wearable device such as a glove or armband to ensure comfort and ease of use. Sensors detect obstacles, motion, and surrounding conditions in real time. The collected data is processed using microcontrollers like ESP32 or Arduino. The processed information is converted into vibration patterns that guide the user. Different vibration intensities represent distance and direction of obstacles. The system operates with low power consumption for daily use. It is designed to be lightweight and affordable. The interface can be customized based on user needs. Overall, the device improves safety, independence, and quality of life for disabled users.





EXISTING SOLUTIONS

1) White Cane (Traditional Blind Stick)
2) Smart Cane with Ultrasonic Sensor
3) Wearable Vibration Bands for Navigation
4) Smartphone-Based Navigation Apps
5) GPS-Based Assistive Devices

CONCLUSION

The proposed haptic interface system provides an affordable and wearable solution for disabled individuals by converting environmental information into tactile feedback. By using embedded platforms like Arduino and ESP32, the system ensures low cost, real-time response, and scalability. This approach improves accessibility, safety, and independence in daily life.

Batch-2:

DEGALA.RAMU	247R1A0579
MOHAMMAD.ABBU	247R1A05A6
G.SURYAPRAKASH	247R1A0585

FACULTY SUPERVISOR

Ms G SWATHI
Assistant professor